

The open systems interconnection osi model is conceptual framework that divides network communications functions into seven layers sending data over a network is complex because various hardware and software technologies must work cohesively

Across geographical and political boundaries the osi data model provides a universal language for computer networking, so diverse technologies can communicate using standard protocols or rules of communication. Every technology in a specific layer must provide certain capabilities and perform specific functions to be useful in networking. Technologies in the higher layers benefit from abstraction as they can use lower-level technologies without having to worry about underlying implementation details

The layers of the Open Systems Interconnection (OSI) model encapsulate every type of network communication across both software and hardware components. The model was designed to allow two standalone systems to communicate via standardised interfaces or protocols based on the current layer of operation.

The benefits of the OSI model are given next.

Engineers can use the OSI model to organize and model complex networked system architectures. They can separate the operating layer of each system component according to its main functionality. The ability to decompose a system into smaller, manageable parts via abstraction makes it easier for people to conceptualize it as a whole

With the OSI reference model, engineers can understand their work better. They know which technological layer (or layers) they're developing for when they create new, networked systems that need to communicate with each other. Engineers can develop networked systems and take advantage of a series of repeatable processes and protocols

The OSI model does not specify the protocols to use between levels, but rather the tasks that protocols perform. It standardizes network communication development so people can rapidly understand, build, and decompose highly complex systems all without prior knowledge of the system. It also abstracts details, so engineers don't require the understanding of every aspect of the model. In modern applications, the lower levels of networking and protocols are abstracted away to simplify system design and development. The following image shows how the OSI model is used in modern application developmentThe Open Systems Interconnection (OSI) model was developed by the International Organization for Standardization and others in the late 1970s. It was published in its first form in 1984 as ISO 7498, with the current version being ISO/IEC 7498-1:1994. The seven layers of the model are given next

The physical layer refers to the physical communication medium and the technologies to transmit data across that medium. At its core, data communication is the transfer of

digital and electronic signals through various physical channels like fiber-optic cables, copper cabling, and air. The physical layer includes standards for technologies and metrics closely related with the channels, such as Bluetooth, NFC, and data transmission speeds

The data link layer refers to the technologies used to connect two machines across a network where the physical layer already exists. It manages data frames, which are digital signals encapsulated into data packets. Flow control and error control of data are often key focuses of the data link layer. Ethernet is an example of a standard at this level. The data link layer is often split into two sub-layers: the Media Access Control (MAC) layer and Logical Link Control (LLC) layer

The network layer is concerned with concepts such as routing, forwarding, and addressing across a dispersed network or multiple connected networks of nodes or machines. The network layer may also manage flow control. Across the internet, the Internet Protocol v4 (IPv4) and IPv6 are used as the main network layer protocols

The primary focus of the transport layer is to ensure that data packets arrive in the right order without losses or errors or can be seamlessly recovered if required. Flow control, along with error control, is often a focus at the transport layer. At this layer, commonly used protocols include the Transmission Control Protocol (TCP), a near-lossless connection-based protocol, and the User Datagram Protocol (UDP), a lossy connectionless protocol. TCP is commonly used where all data must be intact, e.g. file share, whereas UDP is used when retaining all packets is less critical, e.g. video streaming.

The session layer is responsible for network coordination between two separate applications in a session. A session manages the beginning and ending of a one-to-one application connection and synchronization conflicts. Network File System (NFS) and Server Message Block (SMB) are commonly used protocols at the session layer.

The presentation layer is primarily concerned with the syntax of the data itself for applications to send and consume. For example, Hypertext Markup Language (HTML), JavaScript Object Notation (JSON), and Comma Separated Values (CSV) are all modeling data. The application layer is concerned with the specific type of application itself and its standardized communication methods. For example, browsers can communicate using Hypertext Transfer Protocol Secure (HTTPS), and HTTP, and email clients can communicate using POP3 (Post Office Protocol version 3) and SMTP (Simple Mail Transfer Protocol).

Not all systems that use the OSI model implement every layer. Languages to describe the structure of data at the presentation layer

